Score %	IB Grade

EXERCISE 1 [4 marks]

Consider the function

$$f(x) = \frac{3x-5}{x+2}, \qquad x \neq -2.$$

(a) Write down the equations of the vertical and horizontal asymptotes of $y = f(x)$.

(b) Hence write down the value of:

a. $\lim_{x\to\infty} f(x)$

b. $\lim_{x\to -2^+} f(x)$

EXERCISE 2 [4 marks]

Find $\frac{dy}{dx}$ for each of the following:

1. $y = 5x^3 - 4x^2 + 7$

2. $y = 3\sqrt{x} - \frac{2}{x^2}$

3. $y = (2x - 1)(x + 3)$

EXERCISE 3 **[2 marks]**

Use the product rule to find $\frac{dy}{dx}$ if:

$y = (2x + 1)(x^2 - 3)$

EXERCISE 4 **[3 marks]**

Use the quotient rule to find $\frac{dy}{dx}$ if:

$y = \frac{2x - 3}{x + 1}$

EXERCISE 5 **[3 marks]**

1. Use the chain rule to find $\frac{dy}{dx}$ if:

$y = \frac{1}{(3x-2)^3}$

2. Hence, find the gradient of the tangent at $x = -1$.

Answer Key

1. Asymptotes and limits of $f(x) = \dfrac{3x-5}{x+2}$ *[4 marks]*

a. Vertical asymptote: $x = -2$ A1
Horizontal asymptote: $y = 3$ A1
A0 if the vertical asymptote is left as "$x+2=0$" without stating the equation; A0 if the horizontal asymptote is stated without any justification of the ratio of leading coefficients.

b. $\lim\limits_{x\to\infty} f(x) = 3$ A1
$\lim\limits_{x\to -2^+} f(x) = -\infty$ A1
A0 for "undefined" or "does not exist" in place of $-\infty$; the sign must be correct (numerator $\to -11$, denominator $\to 0^+$).

2. Basic differentiation rules *[4 marks]*

a. $y = 5x^3 - 4x^2 + 7 \;\Rightarrow\; \dfrac{dy}{dx} = 15x^2 - 8x$ A1

b. Rewrite as $y = 3x^{1/2} - 2x^{-2}$ M1
$\dfrac{dy}{dx} = \frac{3}{2}x^{-1/2} + 4x^{-3} = \dfrac{3}{2\sqrt{x}} + \dfrac{4}{x^3}$ A1
M0 if index laws applied incorrectly; A1 requires both *terms correct, no partial credit for one correct term.*

c. $y = (2x-1)(x+3) = 2x^2 + 5x - 3 \;\Rightarrow\; \dfrac{dy}{dx} = 4x + 5$ A1
A0 for an answer obtained without first expanding the brackets.

3. Product rule: $y = (2x+1)(x^2-3)$ *[2 marks]*
$u = 2x+1,\ u' = 2; \quad v = x^2-3,\ v' = 2x$
$\dfrac{dy}{dx} = u'v + uv' = 2(x^2-3) + (2x+1)(2x)$ M1
$= 6x^2 + 2x - 6$ A1
M1 for the product-rule structure seen with correct u, u', v, v' (need not be simplified); A1 for full expansion and collection of like terms — an unsimplified but equivalent expression earns M1 only.

4. Quotient rule: $y = \dfrac{2x-3}{x+1}$ *[3 marks]*
$u = 2x-3,\ u' = 2; \quad v = x+1,\ v' = 1$
$\dfrac{dy}{dx} = \dfrac{u'v - uv'}{v^2} = \dfrac{2(x+1) - (2x-3)}{(x+1)^2}$ M1A1
$= \dfrac{5}{(x+1)^2}$ A1
M1 for correct quotient-rule structure with correct u, u', v, v'; first A1 for a correct (even if unsimplified) numerator over $(x+1)^2$; final A1 requires the numerator simplified to 5 — an incorrectly simplified numerator earns A0, not A1FT.

5. Chain rule: $y = \dfrac{1}{(3x-2)^3} = (3x-2)^{-3}$ *[3 marks]*

a. Let $u = 3x-2,\ u' = 3$.

$\dfrac{dy}{dx} = -3u^{-4}\cdot u' = -9(3x-2)^{-4} = \dfrac{-9}{(3x-2)^4}$ M1A1

M1 for correct use of the chain rule (power rule applied to u, multiplied by u'); A1 for the fully correct derivative — no credit if the factor of 3 from u' is omitted or the sign is wrong.

b. At $x=-1$: $3x-2=-5$, so $(3x-2)^4 = 625$.

$\dfrac{dy}{dx} = \dfrac{-9}{625}$ A1FT

Follow-through from the candidate's part (a); award A1FT only if their expression is correctly evaluated at $x=-1$ (not $x=1$).